

Rower Qube

Rower Qube 20251-A0

RP2040 Development Board

Dual-Core ARM Cortex-M0+ Microcontroller With 16 MB

External QSPI Flash Memory

For Embedded Systems, IoT, Robotics, and Educational Applications

Date: August 2025

Revision 1.0

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Description

The ROWER QUBE 20251 is a compact and versatile development board built around the Raspberry Pi RP2040 microcontroller. Designed for embedded systems development, rapid prototyping, and educational applications, the board combines high-performance processing capabilities with a user-friendly hardware architecture. The RP2040 features a dual-core ARM Cortex-M0+ processor operating at up to 133 MHz, providing sufficient computational power for a wide range of real-time control and data-processing applications.

Key Features

- RP2040 Dual-Core ARM Cortex-M0+ Microcontroller
- Up to 133 MHz System Clock Frequency
- 16 MB W25Q128JV QSPI Flash Memory
- USB Programming and Communication Interface
- BOOTSEL Push Button for UF2 Firmware Upload
- 12 MHz Precision Crystal Oscillator
- External 12 V Power Input Support
- 3.3 V LDO Voltage Regulation
- GPIO Expansion Headers
- Power Status LED

Applications

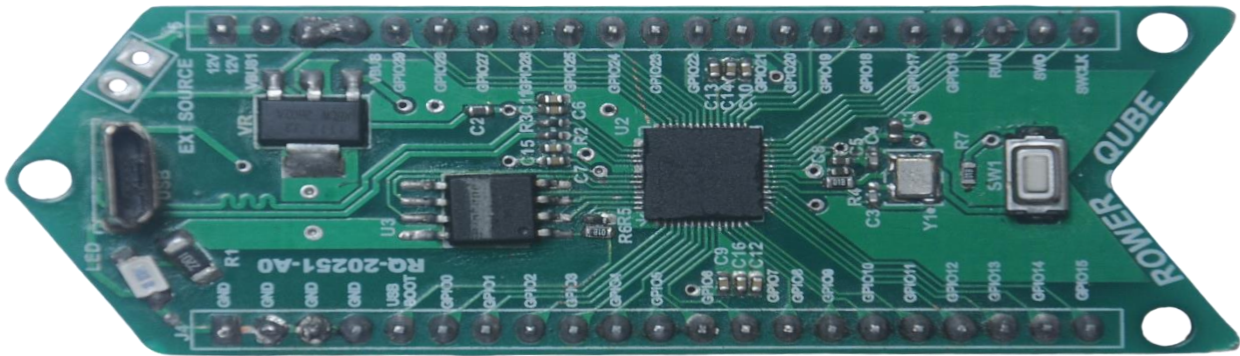
- Embedded Systems Development
- Internet of Things (IoT) Devices
- Robotics and Automation
- Industrial Monitoring Systems
- Sensor Interface Applications
- Educational and Research Projects
- USB Device Development

Mechanical Specifications

Parameter	Value
Board Length	72.8071 mm
Board Width	26.9 mm
Mounting Holes	4
PCB Type	Two-Layer PCB

Product Highlights

- Dual-Core ARM Cortex-M0+ Processor
- Up to 133 MHz Clock Frequency
- 16 MB External QSPI Flash
- USB UF2 Drag-and-Drop Programming
- 12 V External Power Support
- 3.3 V Regulated Operation
- Compact 72.8 mm × 26.9 mm Form Factor



1. Board Overview

1.1 Introduction

The ROWER QUBE 20251 is a custom RP2040-based development board designed for embedded system development, rapid prototyping, and educational applications. The board integrates the Raspberry Pi RP2040 microcontroller with essential support circuitry including external QSPI flash memory, USB connectivity, clock generation circuitry, power management, and GPIO expansion headers.

The RP2040 microcontroller features a dual-core ARM Cortex-M0+ processor operating at frequencies up to 133 MHz. Combined with 16 MB of external flash memory, the board provides a reliable platform for firmware development, sensor interfacing, communication systems, and real-time control applications.

The compact layout and straightforward programming method make the ROWER QUBE 20251 suitable for students, hobbyists, researchers, and embedded system developers seeking a versatile RP2040 development platform.

1.2 Specifications

The ROWER QUBE 20251 is designed to be used as a standalone RP2040 development board for embedded system design, prototyping, and educational applications. The board is based on the Raspberry Pi RP2040 dual-core ARM Cortex-M0+ microcontroller and includes 16 MB of external QSPI flash memory for firmware storage.

Power can be supplied through the onboard Micro USB connector or from an external power source of up to 12 V. A 3.3 V low-dropout voltage regulator provides a stable operating voltage for the microcontroller and supporting circuitry. The board also includes a 12 MHz crystal oscillator for accurate clock generation and a BOOTSEL push button for easy firmware programming through the USB bootloader. GPIO pins are available through external header connectors, allowing straightforward interfacing with sensors, communication modules, displays, and other peripheral devices.

1.3 Device Information

Device Name	Description	Purpose
RP2040	Dual-Core ARM Cortex-M0+ Microcontroller	Main Processing Unit
W25Q128JV	128 Mbit Serial Flash Memory	Firmware and Data Storage
ZM1117-3.3V	Low Dropout Voltage Regulator	3.3 V Power Supply
12 MHz Crystal	Precision Crystal Oscillator	System Clock Source
BOOTSEL Switch	Push Button Switch	USB Bootloader Activation

2. Schematic Diagram

2.1 Overview

The schematic diagram of the ROWER QUBE 20251 development board illustrates the complete hardware implementation of the RP2040-based system. The design integrates the RP2040 microcontroller with supporting circuits including power management, external flash memory, clock generation, USB communication, BOOTSEL functionality, and GPIO expansion headers.

The power section provides regulated 3.3 V operation from either a USB connection or an external power source. The flash memory subsystem provides non-volatile storage for firmware and application data, while the crystal oscillator ensures accurate clock generation for reliable system operation. GPIO header connectors expose the microcontroller pins for interfacing with external peripherals and expansion modules.

The schematic has been designed to provide stable operation, simplified firmware programming, and flexible hardware expansion, making the board suitable for a wide range of embedded system applications.

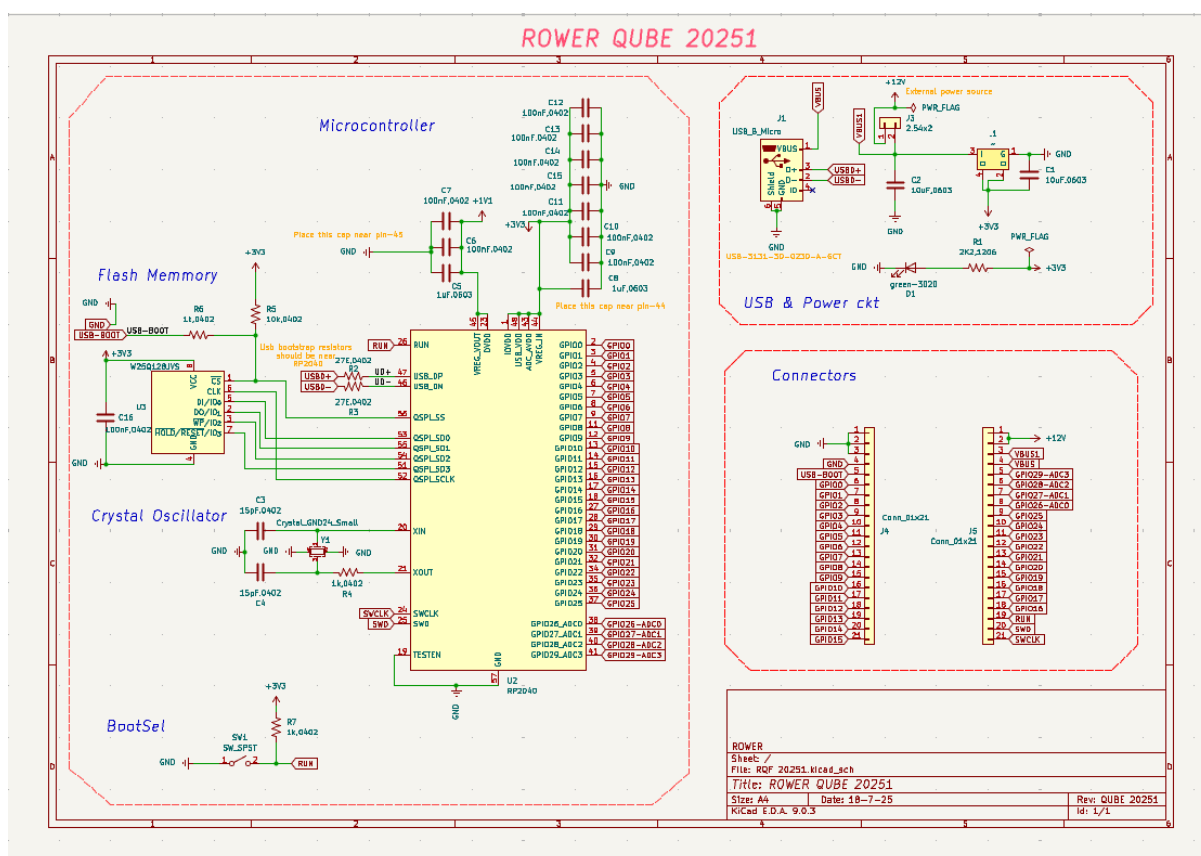


Figure 2-1. Complete Schematic Diagram of ROWER QUBE 20251

3. Hardware Description

3.1 USB and Power Circuit

The USB and Power circuit is designed to provide reliable and flexible power to the ROWER QUBE 20251 development board. The board supports two power input options: a Micro USB connector for programming and standard operation, and an external power input connector capable of accepting voltages up to 12 V.

When powered through the USB connector, a regulated 5 V supply is obtained directly from the host computer or USB adapter. Alternatively, an external power source can be connected through the dedicated power input connector. A Schottky diode is incorporated to provide reverse-polarity protection and prevent damage caused by incorrect power connections.

The selected input voltage is regulated by a ZM1117-3.3 V low-dropout regulator, generating the stable 3.3 V supply required by the RP2040 microcontroller and associated peripherals. Decoupling capacitors are included to suppress noise and improve voltage stability. An onboard LED serves as a power indicator, providing visual confirmation that the board is energized and operating correctly.

In addition to power delivery, the USB interface provides communication and programming functionality through the RP2040 USB subsystem, enabling firmware updates and data exchange over a standard USB connection.

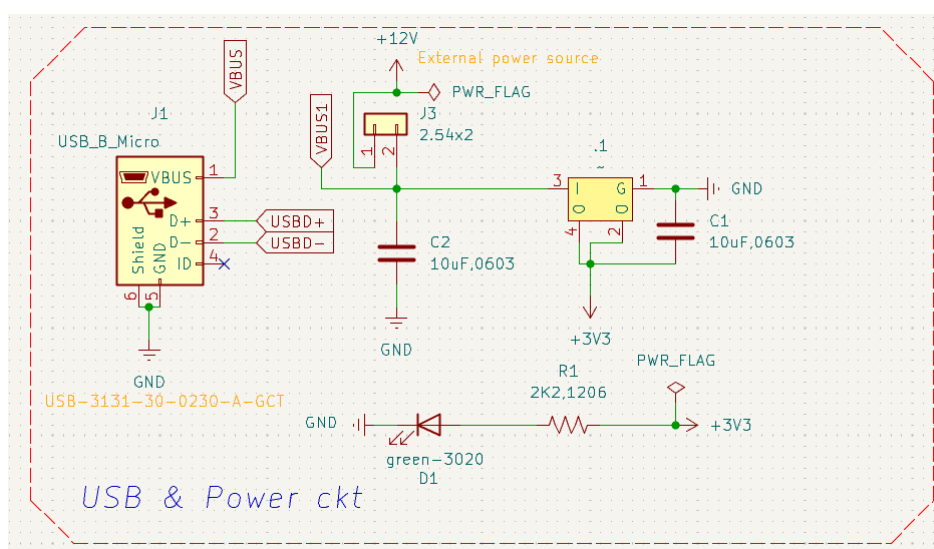


Figure 3-1. USB and Power Circuit Schematic

The crystal oscillator circuit provides the necessary clock signal for the RF2040

TO ensure stable oscillation, two **15 pF capacitors** (C3 and C4) are placed between the

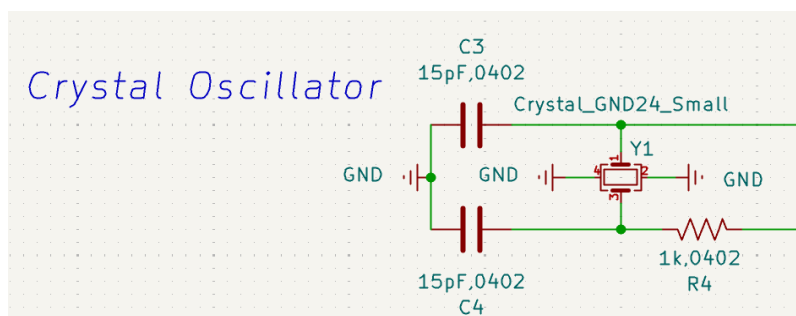


Figure 3-3. Crystal Oscillator Schematic

3.4 Decoupling Capacitors :

Decoupling capacitors play a critical role in stabilizing the power supply across the circuit by filtering out voltage spikes, high-frequency noise, and transient fluctuations. In this schematic, multiple ceramic capacitors typically of values like 0.1 μF , 1 μF , or 10 μF are placed in parallel near the power pins of key components such as the microcontroller (RP2040), voltage regulator, and flash memory. These capacitors act like tiny rechargeable batteries, supplying quick bursts of current to the ICs during switching operations, which helps maintain a stable voltage level.

The reason for using capacitors in parallel is to combine the benefits of different capacitance values. Smaller capacitors (e.g., 0.1 μF) respond faster to high-frequency noise, while larger ones (e.g., 10 μF) are better at handling low-frequency drops or larger current demands. Placing them in parallel ensures a wider range of filtering performance. This design technique is especially important in high-speed digital circuits like the RP2040-based system, where power integrity directly affects stability and reliable operation.

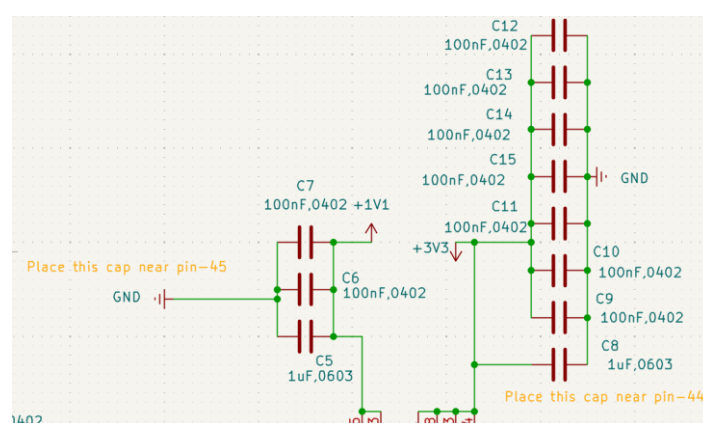


Figure 3-4. Decoupling Capacitors

3.5 Bootsel :

The BOOTSEL (Boot Select) circuit is used to place the RP2040 microcontroller into USB boot mode, allowing the user to upload new firmware without needing external programming tools. This is achieved through a momentary push-button switch connected between the USB_BOOT pin and ground. When this button is pressed and held while the board is powered on or reset, the RP2040 enters USB Mass Storage mode, appearing as a removable drive on the computer. The user can then drag and drop the firmware .uf2 file directly onto the device.

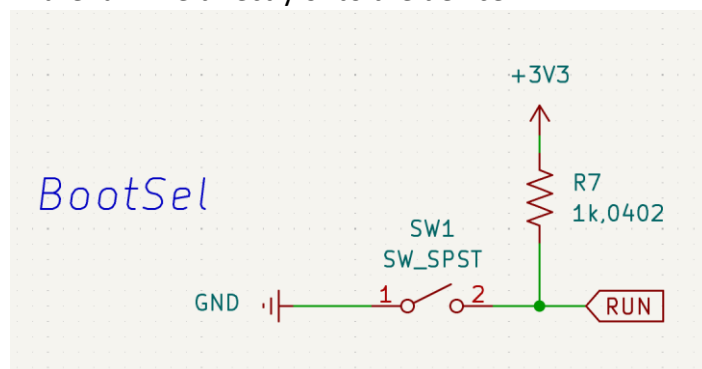


Figure 3-5. Bootsel Schematic

3.6 Connectors:

The schematic includes several connectors that serve as key input/output interfaces for power, programming, and peripheral connectivity. The Micro USB connector (J1) is used for both power supply and USB communication. It allows the board to be powered from a computer or USB charger while also enabling data transfer for programming the RP2040 or interacting with USB-enabled peripherals. The VBUS, USB+, and USB- lines from this connector are routed to both the power management section and the microcontroller's USB interface.

Additionally, the board includes a header connector (J2), typically a pin header, which exposes the GPIO pins of the RP2040 for external interfacing. This allows users to connect sensors, modules, or custom circuits to the microcontroller.

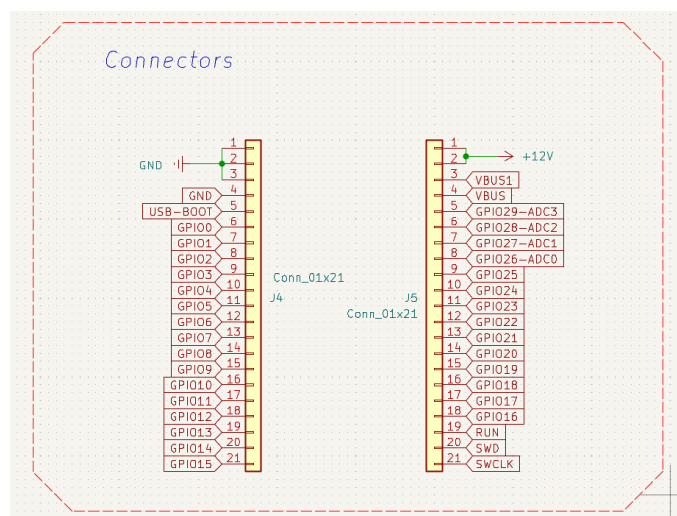


Figure 3-6. Connectors Schematic

4. PCB Layout

4.1 PCB Design Overview

The PCB layout of the ROWER QUBE 20251 development board was designed to provide reliable electrical performance, compact dimensions, and convenient access to all major interfaces. Special attention was given to component placement, power distribution, and signal routing to ensure stable operation of the RP2040 microcontroller and associated circuitry.

The board incorporates dedicated power and ground routing to minimize voltage drops and reduce electrical noise. Critical components such as the crystal oscillator, flash memory, and decoupling capacitors are placed close to the RP2040 microcontroller to improve signal integrity and overall system reliability.

4.2 Top PCB View

The top layer contains the RP2040 microcontroller, flash memory, voltage regulation circuitry, USB interface, and GPIO header connectors. Components are arranged to minimize routing complexity while maintaining accessibility for development and debugging.

4.3 Bottom PCB View

The bottom layer contains routing traces and supporting interconnections required for signal distribution and power delivery. The layout has been optimized to maintain signal integrity and efficient utilization of board space.

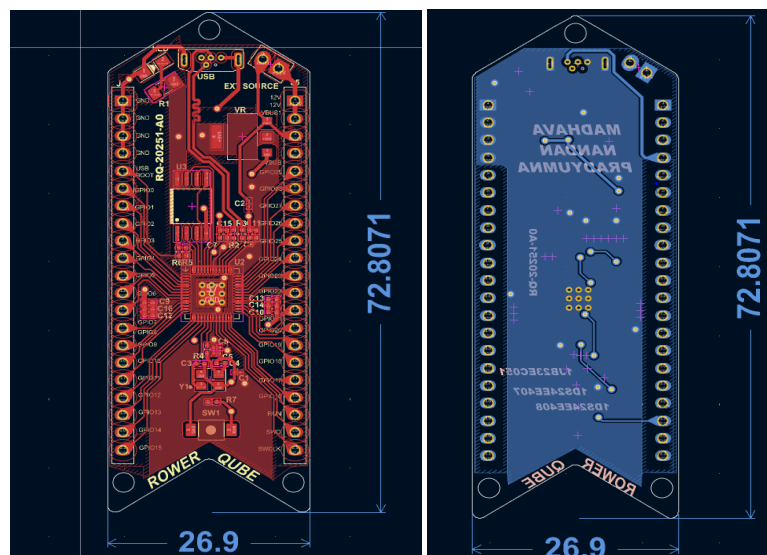


Figure 4-1. Top & Bottom View of the PCB Layout

5. 3D Board View

5.1 3D Visualization

The three-dimensional representation of the ROWER QUBE 20251 development board provides a realistic view of component placement, connector orientation, and overall mechanical structure. The 3D model is generated directly from the PCB design data and serves as a valuable tool for validating component clearances, enclosure compatibility, and assembly requirements before fabrication.

The visualization confirms the placement of the RP2040 microcontroller, flash memory, USB connector, voltage regulator, GPIO headers, and other supporting components. It also assists in identifying potential mechanical constraints and provides a clear representation of the final assembled board.

The compact arrangement of components and connectors contributes to efficient space utilization while maintaining accessibility for programming, power input, and peripheral interfacing.

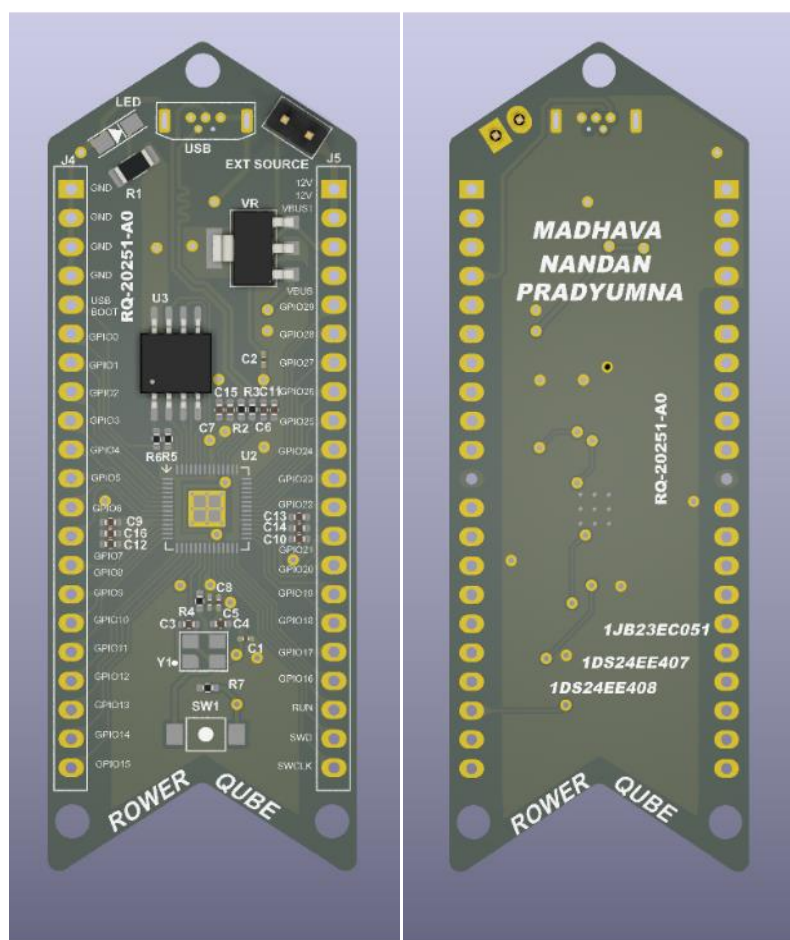


Figure 5-1. 3D View of ROWER QUBE 20251 Development Board

6. Gerber Files

6.1 Gerber Generation

Gerber file generation is a crucial step in the PCB manufacturing process, as it converts the completed PCB design into a standardized format that can be interpreted by fabrication machines. For the ROWER QUBE 20251 Development Board, Gerber files were generated using the KiCad PCB design software after completing the schematic design, PCB layout, and design rule verification. These files serve as the communication medium between the PCB designer and the manufacturing facility.

The generated Gerber package includes all essential layers required for fabrication, such as the top and bottom copper layers, solder mask layers, silkscreen layers, paste layers, board outline, and drill files (Excellon format). Each layer contains precise information regarding track routing, pad locations, vias, component markings, and mechanical dimensions. This ensures that the fabricated board accurately matches the intended design specifications.

Before generating the manufacturing outputs, a Design Rule Check (DRC) was performed in KiCad to identify and eliminate issues such as track clearance violations, unconnected nets, overlapping footprints, and manufacturing constraints. Once the design passed all verification checks, the Gerber files and drill files were exported using KiCad's Plot and Drill Generation tools.

To ensure manufacturing accuracy, the generated files were reviewed using a Gerber viewer. This verification step confirmed proper layer alignment, board dimensions, drill hole placement, and overall layout integrity. The final Gerber package was then compressed and prepared for submission to the PCB fabrication facility. Proper generation and validation of Gerber files significantly reduce fabrication errors, improve production reliability, and ensure successful assembly of the ROWER QUBE 20251 development board.

6.2 Top Copper Layer

The top copper layer contains the primary component footprints, signal traces, power distribution networks, and routing connections required for board functionality. This layer accommodates the RP2040 microcontroller, flash memory device, voltage regulation circuitry, USB interface, and GPIO header connections.

6.3 Bottom Copper Layer

The bottom copper layer provides additional routing resources for signal interconnections and power distribution. The use of a two-layer PCB structure enables efficient routing while maintaining a compact board size and ensuring reliable electrical performance.

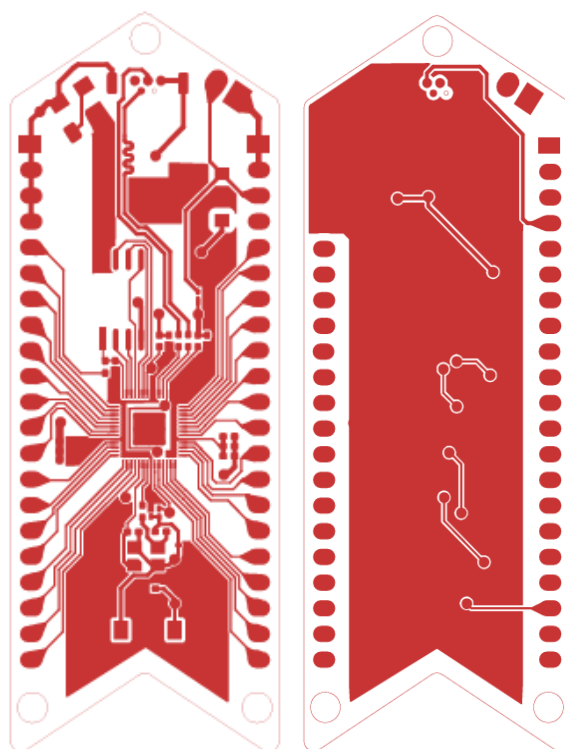


Figure 6-1. Top & Bottom Copper Layer Generated from PCB Design

6.4 Gerber Viewer Output

The generated Gerber files were verified using a Gerber viewing tool prior to fabrication. Verification confirmed correct layer alignment, drill hole placement, board outline definition, silkscreen positioning, and copper trace routing. This process helps identify manufacturing issues before PCB production and ensures the design is fabrication-ready.

After generating the Gerber files, a detailed verification process was carried out using the KiCad Gerber Viewer (GerbView) to ensure that the PCB design was ready for manufacturing. The Gerber Viewer provides a visual representation of all PCB layers and allows designers to inspect the final fabrication data exactly as it will be interpreted by the PCB manufacturer. This verification step is essential for detecting errors that may not be visible during the PCB layout stage.

The board outline and mechanical dimensions were checked to ensure compliance with the intended form factor of the ROWER QUBE 20251 Development Board. Silkscreen markings, component labels, polarity indicators, and reference designators were also reviewed for readability and correct placement. By performing this comprehensive verification, potential fabrication and assembly issues were minimized, ensuring that the PCB design was accurate, reliable, and fully prepared for manufacturing.

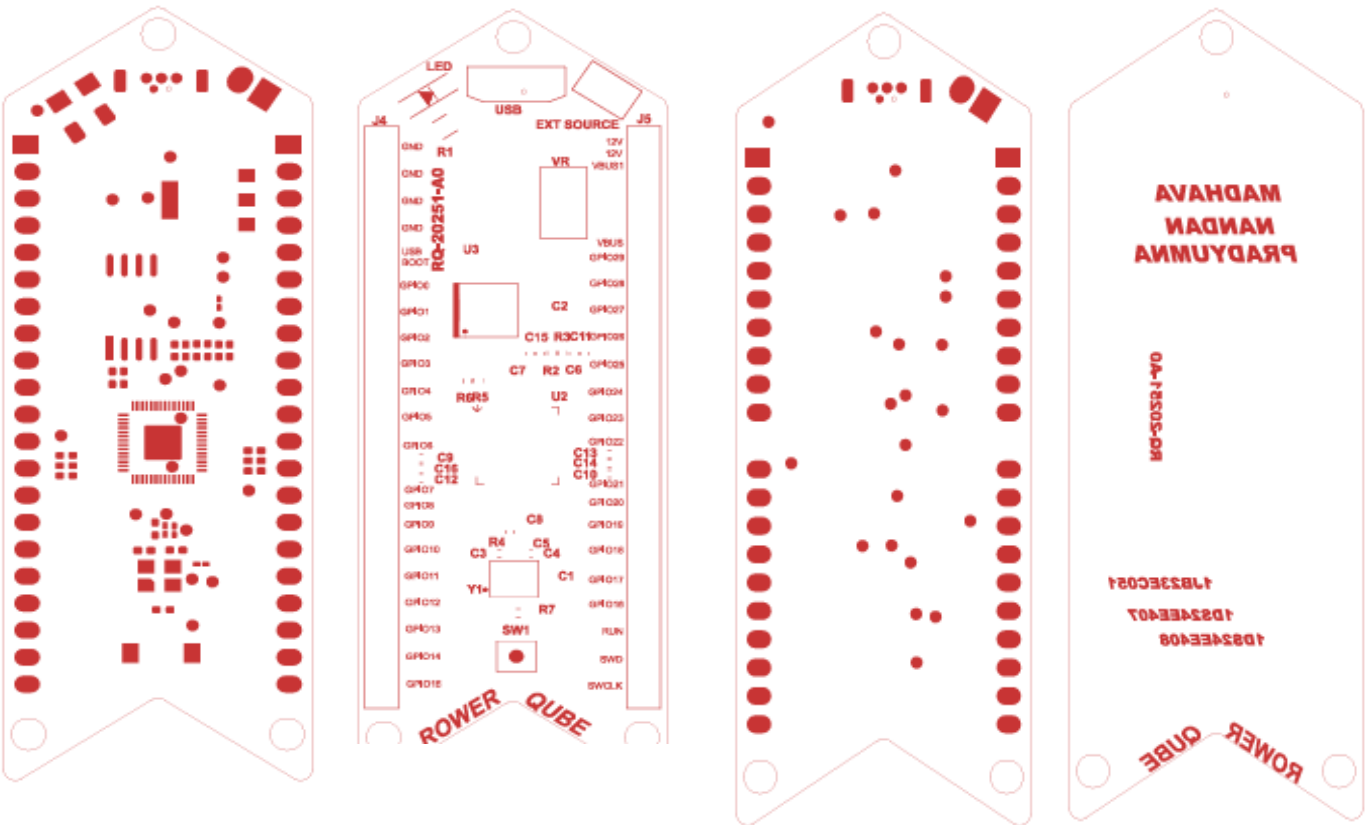


Figure 6-2. Top and Bottom Mask and Silkscreens

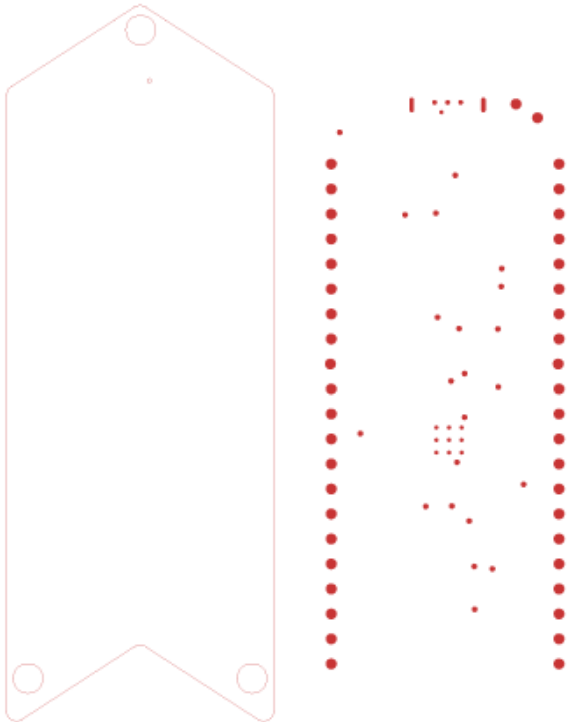


Figure 6-3. Edge Cuts & PTH Drill Holes

7. Bill of Materials (BOM) & Footprints

7.1 Overview

The Bill of Materials (BOM) is a comprehensive list of all electronic and mechanical components required for the fabrication and assembly of the ROWER QUBE 20251 Development Board. It serves as a crucial document that provides detailed information about component specifications, quantities, package types, reference designators, and manufacturer part numbers. The BOM ensures accurate procurement of components and facilitates efficient PCB assembly and maintenance.

The development board incorporates essential components such as the RP2040 microcontroller, voltage regulators, crystal oscillator, flash memory, USB interface components, capacitors, resistors, connectors, and protection devices. Together, these components provide processing capability, power regulation, clock generation, firmware storage, communication interfaces, and system reliability. A well-prepared BOM helps reduce assembly errors, simplifies inventory management, and ensures that all required components are available before manufacturing, thereby supporting the successful development and operation of the ROWER QUBE 20251 platform.

7.2 Bill Of Materials

Ref. Designator	Component	Value / Part Number	Quantity
U2	Microcontroller	RP2040	1
U3	Flash Memory	W25Q128JV	1
U1	Voltage Regulator	ZM1117-3.3V	1
Y1	Crystal Oscillator	12 MHz	1
D1	Power LED	Green LED	1
SW1	BOOTSEL Switch	Push Button	1
J1	USB Connector	Micro USB	1
J4,J5	GPIO Header	Pin Header	2
J3	Power Connector	12 V Input	1
C1,C2	Capacitors	10uF,0603	2
C3,C4	Capacitors	15pF,0402	2
C5,C8	Capacitors	1uF,0603	2
C6,C7,C9 – C16	Capacitors	100nF,0402	10
R1	Resistors	2k2,1206	1
R2,R3	Resistors	27E,0402	2
R4,R6,R7	Resistors	1k,0402	3
R5	Resistors	10k,0402	1

7.3 Footprint & Description

Footprints are the physical PCB representations of electronic components and define the pad layout, dimensions, drill holes, and placement requirements necessary for assembly. In the design of the ROWER QUBE 20251 Development Board, footprints were assigned to all schematic components during the PCB layout stage in KiCad. Most of the required footprints, including passive components, microcontrollers, and other standard devices, were selected from KiCad's built-in footprint libraries.

Ref. Designator	Footprints	Description
U2	Library:QFN-56	A microcontroller by Raspberry Pi
U3	Library:SOIC-8-208mil	128Mbit / 16MiB Serial Flash Memory, Standard/Dual/Quad SPI, 2.7-3.6V, SOIC-8
U1	Library:SOT-223-4	Fixed 3.3 V low-dropout (LDO) voltage Regulator for stable Power Supply
Y1	Library:CRYSTAL-ABM8	Four pin crystal, GND on pins 2 and 4, small symbol
D1	Library:LED-3020	Light emitting diode
SW1	Library:TACT-SW-3X4X2	Single Pole Single Throw (SPST) switch
J1	Library:USB3131-30-0230-A	USB Micro Type B connector
J4,J5	Library:CONN-2X21	Generic connector, single row, 01x21
J3	Library:CONN-2.54X2	Generic connector, single row, 01x02
C1,C2	Capacitor_SMD:C_0201_0603Metric_Pad0.64x0.40mm_HandSolder	Unpolarized capacitor
C3,C4	Capacitor_SMD:C_0402_1005Metric_Pad0.74x0.62mm_HandSolder	Unpolarized capacitor
C5,C8	Capacitor_SMD:C_0201_0603Metric_Pad0.64x0.40mm_HandSolder	Unpolarized capacitor
C6,C7,C9 – C16	Capacitor_SMD:C_0402_1005Metric_Pad0.74x0.62mm_HandSolder	Unpolarized capacitor
R1	Resistor_SMD:R_1206_3216Metric_Pad1.30x1.75mm_HandSolder	Resistor, US symbol
R2,R3	Resistor_SMD:R_0402_1005Metric_Pad0.72x0.64mm_HandSolder	Resistor, US symbol
R4,R6,R7	Resistor_SMD:R_0402_1005Metric_Pad0.72x0.64mm_HandSolder	Resistor, US symbol
R5	Resistor_SMD:R_0402_1005Metric_Pad0.72x0.64mm_HandSolder	Resistor, US symbol

8. Conclusion

8.1 Conclusion

The ROWER QUBE 20251 development board demonstrates the complete design and implementation of a custom RP2040-based embedded platform. The board integrates power management, external flash memory, clock generation circuitry, USB connectivity, and GPIO expansion interfaces into a compact and reliable hardware solution.

The design process included schematic development, PCB layout design, 3D model verification, and Gerber file generation, resulting in a manufacturable and functional development board. The completed hardware platform can be used for embedded system development, educational projects, IoT applications, robotics, and rapid prototyping.

The successful realization of the ROWER QUBE 20251 highlights the practical application of electronic design principles and modern PCB development workflows using KiCad.

8.2 Design Tools

The ROWER QUBE 20251 development board was designed using KiCad EDA software. Schematic capture, PCB layout, 3D visualization, and Gerber generation were performed within the KiCad design environment. All manufacturing outputs were verified before final release.

8.3 Manufacturing Files

The project includes schematic files, PCB layout files, 3D models, drill files, and Gerber manufacturing outputs required for PCB fabrication and assembly.

8.4 Revision History

Revision	Date	Description
Rev 1.0	August 2025	Initial release of ROWER QUBE 20251 development board documentation

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RQ - RP2040